



US012396886B2

(12) **United States Patent**
Sacks et al.

(10) **Patent No.:** **US 12,396,886 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **AUTOMATED CAPSULOTOMY**

(58) **Field of Classification Search**

(71) Applicant: **BELKIN VISION LTD.**, Yavne (IL)

CPC A61F 9/008; A61F 2009/00844; A61F
2009/0087; A61F 2009/00889; A61F
2009/00897

(72) Inventors: **Zachary Shane Sacks**, Modiin (IL);
Michael Belkin, Givat Shmu'el (IL)

(Continued)

(56) **References Cited**

(73) Assignee: **Belkin Vision Ltd.**, Yavne (IL)

U.S. PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 268 days.

2,635,502 A 4/1953 Richards
3,594,072 A 7/1971 Feather
(Continued)

(21) Appl. No.: **17/926,633**

FOREIGN PATENT DOCUMENTS

(22) PCT Filed: **May 16, 2021**

AU 2015210430 A1 9/2015
AU 2015315113 B2 3/2016

(86) PCT No.: **PCT/IB2021/054187**

(Continued)

§ 371 (c)(1),

(2) Date: **Nov. 21, 2022**

OTHER PUBLICATIONS

CN Application # 2020800163407 Office Action dated Feb. 4, 2023.

(Continued)

(87) PCT Pub. No.: **WO2022/018525**

PCT Pub. Date: **Jan. 27, 2022**

Primary Examiner — Aaron F Roane

(74) Attorney, Agent, or Firm — Meitar Patents Ltd.;
Daniel Kligler

(65) **Prior Publication Data**

US 2023/0201034 A1 Jun. 29, 2023

(57) **ABSTRACT**

A system (20) includes a radiation source (48) and a controller (44). The controller is configured to define a treatment zone (88) on a capsule (86) of an eye (25) of a subject (22), and to form an opening (96) in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions (94) within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process, acquiring an image (98) of at least part of the capsule, designating one of the target regions based on the acquired image, and causing the radiation source to irradiate the designated target region. Other embodiments are also described.

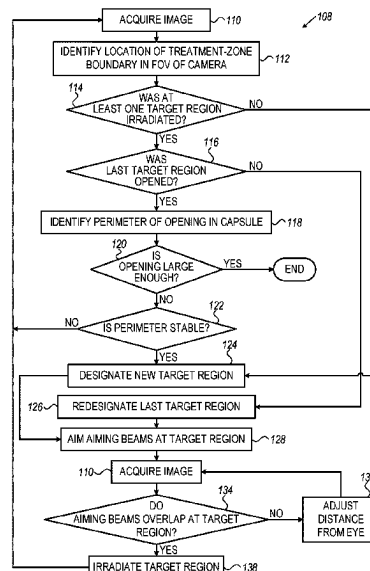
Related U.S. Application Data

(60) Provisional application No. 63/053,650, filed on Jul. 19, 2020.

(51) **Int. Cl.**
A61F 9/008 (2006.01)

(52) **U.S. Cl.**
CPC **A61F 9/008** (2013.01); **A61F 2009/00844**
(2013.01); **A61F 2009/0087** (2013.01);
(Continued)

29 Claims, 7 Drawing Sheets



(52)	U.S. CL.	CPC <i>A61F 2009/00889</i> (2013.01); <i>A61F 2009/00897</i> (2013.01)	8,475,433 B2	7/2013	Mrochen et al.
			8,545,020 B2	10/2013	Liesfeld et al.
(58)	Field of Classification Search	USPC 606/4 See application file for complete search history.	8,568,393 B2	10/2013	Palanker
			8,630,388 B2	1/2014	Gertner et al.
(56)	References Cited	U.S. PATENT DOCUMENTS	8,679,100 B2	3/2014	Raksi et al.
			8,708,491 B2	4/2014	Frey et al.
			8,709,029 B2	4/2014	Griffis, III et al.
			8,771,261 B2	7/2014	Andersen et al.
			8,811,657 B2	8/2014	Teiwes et al.
			8,845,625 B2	9/2014	Angeley et al.
			8,903,468 B2	12/2014	Peyman
			8,920,407 B2	12/2014	Raksi et al.
			8,939,965 B2	1/2015	Liesfeld et al.
			8,968,279 B2	3/2015	Arnoldussen
			8,995,618 B2	3/2015	Gertner
			9,055,896 B2	6/2015	Amthor et al.
			9,192,780 B2	11/2015	McDaniel
			9,220,407 B2	12/2015	Yam et al.
			9,351,878 B2	5/2016	Muehlhoff et al.
			9,480,599 B2	11/2016	Degani et al.
			9,495,743 B2	11/2016	Angeley et al.
			9,504,609 B2	11/2016	Kurtz
			9,532,712 B2	1/2017	Liesfeld et al.
			9,622,911 B2	4/2017	Rubinfeld et al.
			9,782,232 B1	10/2017	Papac
			9,849,032 B2	12/2017	Schuele et al.
			9,849,034 B2	12/2017	Artsyukhovich et al.
			9,877,633 B2	1/2018	Zhao et al.
			9,889,043 B2	2/2018	Frey et al.
			9,968,483 B2	5/2018	Takeda et al.
			10,022,457 B2	7/2018	Peyman
			10,064,757 B2	9/2018	Berlin
			10,143,590 B2	12/2018	Dick et al.
			10,244,991 B2	4/2019	Shademan et al.
			10,258,507 B2	4/2019	Gonzalez et al.
			10,278,865 B2	5/2019	Luttrull et al.
			10,299,961 B2	5/2019	Luttrull et al.
			10,363,169 B2	7/2019	Belkin et al.
			10,441,465 B2	10/2019	Hart et al.
			10,449,091 B2	10/2019	Angeley et al.
			10,456,209 B2	10/2019	Peyman
			10,478,342 B2	11/2019	Dick et al.
			10,524,656 B2	1/2020	Wiltberger et al.
			10,617,564 B1	4/2020	Andersen et al.
			10,684,449 B2	6/2020	Curatu et al.
			10,702,416 B2	7/2020	Belkin et al.
			10,849,789 B2	12/2020	Dewey et al.
			10,925,768 B2	2/2021	Charles
			11,406,536 B2	8/2022	Schuele et al.
			2001/0027314 A1	10/2001	Peyman
			2002/0013572 A1	1/2002	Berlin
			2002/0013573 A1	1/2002	Telfair et al.
			2002/0026179 A1	2/2002	Toh
			2003/0179344 A1	9/2003	Van de Velde
			2003/0225398 A1	12/2003	Zepkin et al.
			2004/0039378 A1	2/2004	Lin
			2004/0059321 A1	3/2004	Knopp et al.
			2004/0196431 A1	10/2004	Farberov
			2005/0096639 A1	5/2005	Slatkine et al.
			2005/0107774 A1	5/2005	Lin
			2005/0185138 A1	8/2005	Wong et al.
			2005/0197655 A1	9/2005	Telfair et al.
			2005/0254009 A1	11/2005	Baker et al.
			2005/0286019 A1	12/2005	Wiltberger et al.
			2005/0288745 A1	12/2005	Andersen et al.
			2006/0100677 A1	5/2006	Blumenkranz et al.
			2006/0176913 A1	8/2006	Souhaite et al.
			2006/0195076 A1	8/2006	Blumenkranz et al.
			2006/0224147 A1	10/2006	Abe et al.
			2006/0265030 A1	11/2006	McDaniel
			2007/0081166 A1	4/2007	Brown et al.
			2007/0129709 A1	6/2007	Andersen et al.
			2007/0159600 A1	7/2007	Gil et al.
			2007/0213693 A1	9/2007	Plunkett
			2008/0027418 A1	1/2008	Berry
			2008/0089481 A1	4/2008	Gertner
			2008/0108934 A1	5/2008	Berlin et al.
			2008/0161781 A1	7/2008	McCardle et al.
			2008/0167642 A1	7/2008	Palanker et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2008/0204658 A1 8/2008 Van Saarloos
 2008/0234667 A1 9/2008 Lang et al.
 2008/0255546 A1 10/2008 Orbachovski
 2009/0137993 A1 5/2009 Kurtz
 2009/0157062 A1 6/2009 Hauger et al.
 2009/0247997 A1 10/2009 Watanabe et al.
 2010/0002837 A1 1/2010 Gertner et al.
 2010/0057059 A1 3/2010 Makino
 2010/0076419 A1 3/2010 Chew et al.
 2010/0142767 A1 6/2010 Fleming
 2010/0324543 A1 12/2010 Kurtz et al.
 2011/0144627 A1 6/2011 Smith et al.
 2011/0172649 A1 7/2011 Schuele et al.
 2011/0190741 A1 8/2011 Deisinger et al.
 2012/0016349 A1 1/2012 Brownell
 2012/0050308 A1 3/2012 Nakano et al.
 2012/0083772 A1 4/2012 Rubinfeld et al.
 2012/0089134 A1 4/2012 Horvath et al.
 2012/0259321 A1 10/2012 Vera et al.
 2012/0283557 A1 11/2012 Berlin
 2013/0103011 A1 4/2013 Grant et al.
 2013/0123761 A1 5/2013 Belkin et al.
 2013/0158530 A1 6/2013 Goldshleger et al.
 2013/0204236 A1 8/2013 Awdeh
 2013/0218145 A1 8/2013 Belkin et al.
 2013/0289450 A1 10/2013 Homer
 2013/0317570 A1 11/2013 Luttrull et al.
 2014/0094785 A1 4/2014 Charles
 2014/0114297 A1 4/2014 Woodley et al.
 2014/0128731 A1 5/2014 Gonzalez et al.
 2014/0128851 A1 5/2014 Wysopal
 2014/0128852 A1 5/2014 Gooding et al.
 2014/0135747 A1 5/2014 Donitzky et al.
 2014/0135753 A1 5/2014 Feklistov et al.
 2014/0276681 A1 9/2014 Schuele et al.
 2014/0307077 A1 10/2014 Prabhakar
 2015/0164635 A1 6/2015 Renke
 2015/0190276 A1 7/2015 Ha et al.
 2015/0223683 A1 8/2015 Davidovics et al.
 2015/0266706 A1 9/2015 Hashimoto
 2015/0272782 A1 10/2015 Schuele et al.
 2015/0313759 A1 11/2015 Vera et al.
 2016/0008169 A1 1/2016 Yu
 2016/0008172 A1 1/2016 Kahook
 2016/0008923 A1 1/2016 Astle et al.
 2016/0067035 A1 3/2016 Gontijo et al.
 2016/0067087 A1 3/2016 Tedford et al.
 2016/0089232 A1 3/2016 DeBoer et al.
 2016/0089269 A1 3/2016 Horvath et al.
 2016/0095752 A1 4/2016 Srinivasan et al.
 2016/0113816 A1 4/2016 Herekar et al.
 2016/0346126 A1 12/2016 Luttrull et al.
 2016/0354241 A1 12/2016 Mordaunt et al.
 2016/0367399 A1 12/2016 Goldshleger et al.
 2017/0000647 A1 1/2017 Schuele et al.
 2017/0038284 A1 2/2017 Nemati
 2017/0087014 A1 3/2017 Potter, Jr. et al.
 2017/0112663 A1 4/2017 Bareket et al.
 2017/0127938 A1 5/2017 Izatt et al.
 2017/0184875 A1 6/2017 Newman
 2017/0246033 A1 8/2017 Bor et al.
 2017/0340483 A1 11/2017 Rill et al.
 2017/0360604 A1 12/2017 Bach et al.
 2018/0085257 A1 3/2018 Horvath et al.
 2018/0104477 A1 4/2018 Kurtz et al.
 2018/0125708 A1 5/2018 Bohme et al.
 2018/0168737 A1 6/2018 Ren et al.
 2018/0207029 A1 7/2018 Herekar et al.
 2018/0214305 A1 8/2018 Schuele et al.
 2018/0221199 A1 8/2018 Heacock
 2018/0235462 A1 8/2018 Gooi et al.
 2018/0344527 A1 12/2018 Palanker et al.
 2019/0078073 A1 3/2019 Streeter et al.
 2019/0099291 A1 4/2019 Herekar et al.
 2019/0105200 A1 4/2019 Hipsley

2019/0105519 A1 4/2019 Herekar et al.
 2019/0117459 A1 4/2019 Berlin
 2019/0142636 A1 5/2019 Tedford et al.
 2019/0151146 A1 5/2019 Kim
 2019/0247225 A1 8/2019 Stobrawa et al.
 2019/0269554 A1 9/2019 Goldshleger et al.
 2019/0343680 A1 11/2019 Belkin et al.
 2019/0344076 A1 11/2019 Irazoqui et al.
 2019/0358085 A1 11/2019 Fu et al.
 2020/0038245 A1 2/2020 Hart et al.
 2020/0078216 A1 3/2020 Raksi
 2020/0093639 A1 3/2020 McCall, Jr.
 2020/0107724 A1 4/2020 Wiltberger et al.
 2020/0146887 A1 5/2020 Horvath et al.
 2020/0306080 A1 10/2020 Herekar et al.
 2020/0345546 A1 11/2020 Belkin et al.
 2020/0352785 A1 11/2020 Holland et al.
 2020/0360187 A1 11/2020 Schuele et al.
 2020/0379216 A1 12/2020 Curatu et al.
 2021/0113373 A1 4/2021 Sacks et al.
 2021/0267800 A1 9/2021 Sacks et al.
 2021/0338484 A1 11/2021 Hipsley
 2022/0031503 A1* 2/2022 Dorin A61F 9/008
 2022/0249861 A1 8/2022 Belkin et al.
 2023/0201037 A1 6/2023 Barrett et al.
 2023/0226372 A1 7/2023 Herekar et al.

FOREIGN PATENT DOCUMENTS

CA 2640203 A1 8/2007
 CN 1579351 A 2/2005
 CN 101411607 A 4/2009
 CN 201537172 U 8/2010
 CN 102193182 A 9/2011
 CN 105138996 A 12/2015
 CN 205698218 U 11/2016
 CN 108024870 A 5/2018
 DE 202016006265 U1 3/2017
 EP 0224322 A1 6/1987
 EP 0651982 A1 5/1995
 EP 0689811 A1 1/1996
 EP 1602321 A1 12/2005
 EP 2301421 A1 3/2011
 EP 2301424 B1 3/2011
 EP 2301425 B1 3/2011
 EP 2602005 A1 6/2013
 EP 1856774 B1 6/2016
 EP 2695016 B1 3/2017
 EP 2992931 B1 8/2017
 EP 2391318 B1 12/2017
 EP 3329839 A1 6/2018
 EP 2729099 B1 11/2019
 EP 3191040 B1 7/2020
 EP 3517081 B1 11/2020
 EP 2854729 B1 3/2021
 FR 2655837 A1 6/1991
 JP 2007151739 A 6/2007
 JP 2010148635 A 7/2010
 JP 2016013255 A 1/2016
 JP 2018051210 A 4/2018
 KR 20180106113 A 10/2018
 KR 20190022216 A 3/2019
 RU 2499582 C1 11/2013
 RU 2553507 C1 6/2015
 WO 9216259 A1 10/1992
 WO 1993012727 A1 7/1993
 WO 9316631 A1 9/1993
 WO 9412092 A1 6/1994
 WO 9416425 A1 7/1994
 WO 9515134 A1 6/1995
 WO 1998022016 A2 5/1998
 WO 9918868 A1 4/1999
 WO 0195842 A1 12/2001
 WO 02064031 A2 8/2002
 WO 02087442 A1 11/2002
 WO 2014018104 A1 1/2004
 WO 2004027487 A1 4/2004
 WO 2006119349 A2 11/2006
 WO 2006119584 A1 11/2006

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2006128038	A2	11/2006
WO	2007103349	A2	9/2007
WO	2008112236	A1	9/2008
WO	2008118198	A2	10/2008
WO	2010094353	A1	8/2010
WO	2010113193	A1	10/2010
WO	2011017002	A2	2/2011
WO	2011163508	A2	6/2011
WO	2011085274	A1	7/2011
WO	2011151812	A1	12/2011
WO	2013004255	A1	1/2013
WO	2013035091	A1	3/2013
WO	2013059481	A1	4/2013
WO	2013059564	A1	4/2013
WO	2013122711	A1	8/2013
WO	2013165689	A1	11/2013
WO	2014025862	A1	2/2014
WO	2014132162	A1	9/2014
WO	2014191031	A1	12/2014
WO	2015069197	A1	5/2015
WO	2015119888	A1	8/2015
WO	2015130821	A2	9/2015
WO	2015131135	A1	9/2015
WO	2016018864	A1	2/2016
WO	2016058931	A2	4/2016
WO	2016156760	A1	10/2016
WO	2016187436	A1	11/2016
WO	2016207739	A1	12/2016
WO	2017023296	A1	2/2017
WO	2017031570	A1	3/2017
WO	2017069819	A1	4/2017
WO	2018005796	A1	1/2018
WO	2018021780	A1	2/2018
WO	2018049246	A1	3/2018
WO	2018152020	A1	8/2018
WO	2018232397	A1	12/2018
WO	2019109125	A1	6/2019
WO	2020008323	A1	1/2020
WO	2020012841	A1	1/2020
WO	2020018242	A1	1/2020
WO	2020018436	A1	1/2020
WO	2020050308	A1	3/2020
WO	202093060	A2	5/2020
WO	2020089737	A1	5/2020
WO	2020093060	A2	5/2020
WO	2020183342	A1	9/2020
WO	2021026538	A1	2/2021
WO	2021048723	A1	3/2021
WO	2021155445	A1	8/2021
WO	2021170664	A1	9/2021
WO	2022223690	A1	10/2022

OTHER PUBLICATIONS

JP Application # 2020561860 Office Action dated Feb. 7, 2023.

Kohnen et al., "Internal Anterior Chamber Diameter using Optical Coherence Tomography Compared with White-To-White Distances using Automated Measurements," *Journal of Cataract & Refractive Surgery*, vol. 32, pp. 1809-1813, Nov. 2006.

Zhang et al., "Perioperative Medications for Preventing Temporarily Increased Intraocular Pressure after Laser Trabeculoplasty (Review)," *Cochrane Database of Systematic Reviews* 2017, issue 2, pp. 1-117, year 2017.

Katta et al., "Optical Coherence Tomography Image-Guided Smart Laser Knife for Surgery," *Lasers in Surgery and Medicine*, Wiley Online Library, pp. 1-11, Jul. 2017.

Barnes et al., "Control of Intraocular Pressure Elevations after Argon Laser Trabeculoplasty: Comparison of Brimonidine 0.2% to Apraclonidine 1.0%," *Ophthalmology*, vol. 106, No. 10, pp. 2033-2037, year 1999.

Yakopson et al., "Brimonidine 0.1% vs. Apraclonidine 0.5% for Prevention of Intraocular Pressure Elevation after Selective Laser Trabeculoplasty," *Investigative Ophthalmology & Visual Science*, vol. 49, p. 1234, May 2008.

Kim et al., "Effect of Prophylactic Topical Brimonidine (0.15%) Administration on the Development of Subconjunctival Hemorrhage after Intravitreal Injection," *Retina, The Journal for Retinal and Vitreous Diseases*, vol. 31, No. 2, pp. 389-392, year 2011.

Hong et al., "Effect of Prophylactic Brimonidine Instillation on Bleeding during Strabismus Surgery in Adults," *American Journal of Ophthalmology*, vol. 144, No. 3, pp. 469-470, Sep. 2007.

Goldsmith et al., "Anterior Chamber Width Measurement by High-Speed Optical Coherence Tomography," *Ophthalmology*, vol. 112, No. 2, pp. 238-244, year 2005.

Norden, "Effect of Prophylactic Brimonidine on Bleeding Complications and Flap Adherence After Laser in situ Keratomileusis," *Journal of Refractive Surgery*, vol. 18, No. 4, pp. 468-471, Jul./Aug. 2002.

Gophotonics, "NL200 series," Data Sheet, pp. 1-3, Jun. 29, 2017.

EP Application # 20201567.3 Office Action dated Jun. 6, 2023.

JP Application # 2020561860 Office Action dated Jun. 13, 2023.

JP Application # 2021516473 Office Action dated Jun. 20, 2023.

CN Application # 2020800563096 Office Action dated Jul. 1, 2023.

EP Application # 20864109.2 Search Report dated Aug. 10, 2023.

Gazzard et al., "Selective Laser Trabeculoplasty versus Drops for Newly Diagnosed Ocular Hypertension and Glaucoma: The LIGHT RCT," *Health Technology Assessment, NHS*, vol. 23, issue 31, pp. 1-132, Jun. 2019.

Kelley et al., "Stem Cells in the Trabecular Meshwork: Present and Future Promises," *Experimental Eye Research*, vol. 88, issue 4, pp. 747-751, Apr. 2009.

Dueker et al., "Stimulation of Cell Division by Argon and Nd:YAG Laser Trabeculoplasty in Cynomolgus Monkeys," *Investigative Ophthalmology & Visual Science*, vol. 31, No. 1, pp. 115-124, year 1990.

Nowell et al., "Corneal Epithelial Stem Cells and their Niche at a Glance," *Cell Science at a Glance*, vol. 130, Issue 6, pp. 1021-1025, year 2017.

Kim et al., "Diagnosis of Corneal Limbal Stem Cell Deficiency," *Current Opinion in Ophthalmology*, Wolters Kluwer Health, Inc., vol. 28, No. 4, pp. 355-362, Jul. 2017.

Gonzalez et al., "Limbal Stem Cells: Identity, Developmental Origin, and Therapeutic Potential," *WIREs Developmental Biology*, Wiley, vol. 7, issue 2, pp. 1-23, Mar. 2018.

Sepehr, "Corneal Endothelial Cell Dysfunction: Etiologies and Management," *Therapeutic Advances in Ophthalmology*, pp. 1-19, year 2018.

Walshe et al., "Serial Explant Culture Provides Novel Insights into the Potential Location and Phenotype of Corneal Endothelial Progenitor Cells," *Experimental Eye Research*, vol. 127, pp. 9-13, year 2014.

Pinnamaneni et al., "Concise Review: Stem Cells in the Corneal Stroma," *Stem Cells*, vol. 30, issue 6, pp. 1059-1063, year 2012.

Espana et al., "Existence of Corneal Endothelial Slow-Cycling Cells," *Investigative Ophthalmology & Visual Science*, vol. 56, No. 6, pp. 3827-3837, Jun. 2015.

EP Application # 21885460.2 Search Report dated Aug. 26, 2024.

EP Application # 19830473.5 Office Action dated Sep. 3, 2024.

U.S. Appl. No. 17/273,323 Office Action dated Oct. 30, 2024.

U.S. Appl. No. 17/427,926 Office Action dated Aug. 27, 2024.

U.S. Appl. No. 17/427,926 Office Action dated May 9, 2024.

EP Application # 19877990.2 Office Action dated May 13, 2024.

EP Application # 24158977.9 Search Report dated May 15, 2024.

EP Application # 21845437.9 Search Report dated Jun. 19, 2024.

JP Application # 2023217477 Office Action dated Jul. 9, 2024.

U.S. Appl. No. 17/273,323 Office Action dated Jun. 18, 2024.

Nagar et al., "A randomised, prospective study comparing selective laser trabeculoplasty with latanoprost for the control of intraocular pressure in ocular hypertension and open angle glaucoma," *British Journal of Ophthalmology*, vol. 89, pp. 1413-1417, year 2005.

Hong et al., "Repeat Selective Laser Trabeculoplasty," *Journal of Glaucoma*, vol. 18, issue 3, pp. 180-183, Mar. 2009.

Goyal et al., "Effect of primary selective laser trabeculoplasty on topographic outflow facility—a randomised clinical trial," *British Journal of Ophthalmology*, BMJ Publishing Group, vol. 94, issue 11, pp. 1-22, year 2010.

(56)

References Cited**OTHER PUBLICATIONS**

- Franco et al., "Effect of Second SLT on IOP," *Investigative Ophthalmology & Visual Science*, vol. 48, pp. 1-2, May 2007.
- Chen et al., "A Comparison between 90 degrees and 180 degrees Selective Laser Trabeculoplasty," *Journal of Glaucoma*, vol. 13, issue 1, p. 1, Feb. 2004.
- Mequio et al., "Efficacy of Repeat Selective Laser Trabeculoplasty," *Investigative Ophthalmology & Visual Science*, vol. 48, p. 1, year 2007.
- Grulkowski et al., "Anterior segment imaging with Spectral OCT system using a high-speed CMOS camera," *Optics Express*, vol. 17, No. 6, p. 4842-4858, year 2009.
- Shields et al., "Noncontact Transscleral Nd:YAG Cyclophotocoagulation: A Long-Term Follow-Up of 500 Patients," *Transactions of the American Ophthalmological Society*, vol. XCII, pp. 271-287, year 1994.
- Liu et al., "Real-time visual analysis of microvascular blood flow for critical care," CVPR2015 paper as Open Access Version, provided by the Computer Vision Foundation, pp. 2217-2225, year 2015.
- Desco et al., "Effect of prophylactic brimonidine on bleeding complications after cataract surgery," *European Journal of Ophthalmology*, vol. 15, pp. 228-232, year 2005.
- Pasquali et al., "Dilute brimonidine to improve patient comfort and subconjunctival hemorrhage after LASIK," *Journal of Refractive Surgery*, vol. 29, pp. 469-475, year 2013.
- Sacks et al., "Non-contact direct selective laser trabeculoplasty: light propagation analysis," *Biomedical Optics Express*, vol. 11, pp. 2889-2904, year 2020.
- Kasuga et al., "Trabecular Meshwork Length in Men and Women by Histological Assessment," *Current Eye Research*, Early Online, pp. 1-5, Jun. 2012.
- Navilas Operator Manual, Document Version 2.10, 2012 OD-OS GmbH, pp. 1-94, Sep. 2012.
- SensoMotoric Instruments GmbH (SMI), "SG 3000", Product Flyer, pp. 1-2, year 2010.
- Ashik et al., "The precision of ophthalmic biometry using calipers," *Canadian Journal of Ophthalmology*, vol. 48, Issue 6, pp. 1-13, Dec. 2013.
- Balalzi, "Noncontact Thermal Mode Nd:YAG Laser Transscleral Cyclocoagulation in the Treatment of Glaucoma," *Ophthalmology*, vol. 98, pp. 1858-1863, year 1991.
- Leung et al., "Anterior chamber angle imaging with optical coherence tomography," *Eye*, vol. 25, pp. 261-267, year 2011.
- Tasman et al., "The Wills Eye Hospital Atlas of Clinical Ophthalmology," Lippincott Williams & Wilkins, p. 158, year 2001.
- Gaasterland, "Laser Therapies: Iridotomy, Iridoplasty, and Trabeculoplasty," as appears in "The Glaucoma Book: A Practical Evidence-Based Approach to Patient Care," Springer, p. 722, year 2010.
- Kara, "Bleeding in Retinal Images Using Image Processing," A Thesis submitted to the graduate school of applied sciences of Near East University, Nicosia, Larnaca, pp. 1-79, year 2019.
- CN Application # 201980070459X Office Action dated Dec. 23, 2022.
- "Smart Selecta Duet—Your Smart Selection for Glaucoma Care," Product Brochure, pp. 1-6, The Lumenis Group of Companies, year 2018.
- Rashad, "How to do YAG Laser Posterior Capsulotomy in Small Pupil?," YouTube Clip, p. 1, Jul. 19, 2020 <https://www.youtube.com/watch?v=12c39BoNBjM>.
- Quantel Medical, "Optimis II", datasheet, pp. 1-4, Aug. 5, 2022.
- Variscite, "DART-MX8M", Datasheet, pp. 1-92, Feb. 2021.
- Rashad, "What is the Easiest Way to do YAG Laser Posterior Capsulotomy?," YouTube Clip, p. 1, Jul. 17, 2020 https://www.youtube.com/watch?v=g0pV3UGo_20.
- International Application PCT/IB2021/054187 search report dated Jul. 30, 2021.
- Vogel et al., "Optical properties of human sclera, and their consequences for transscleral laser applications," *Lasers In Surgery and Medicine*, vol. 11, pp. 331-340, year 1991.
- Geffen et al., "Transscleral Selective Laser Trabeculoplasty Without a Gonioscopy Lens", *Journal of Glaucoma*, Inc, vol. 26, No. 3, pp. 201-207, Mar. 2017.
- Das et al., "Sclera Recognition—A Survey", 2nd IAPR Asian Conference on Pattern Recognition, pp. 1-5, year 2013.
- Kaya et al., "Designing A Pattern Stabilization Method Using Scleral Blood Vessels For Laser Eye Surgery", International Conference on Pattern Recognition, pp. 698-701, Istanbul, Turkey, Aug. 23-26, year 2010.
- Barkana et al., "Selective Laser Trabeculoplasty", *Survey of Ophthalmology*, vol. 52, No. 6, pp. 634-653, year 2007.
- Arany, "Photobiomodulation therapy: Easy to do, but difficult to get right", *LaserFocusWorld*, pp. 1-6, Jul. 31, 2019 downloaded from www.laserfocusworld.com/lasers-sources/article/14037967/photobiomodulation-therapyeasy-to-do-put-difficult-to-get-right, pp. 22-24, year 2019.
- Borzabadi-Farahani, "Effect of low-level laser irradiation on proliferation of human dental mesenchymal stem cells; a systemic review", *Journal of Photochemistry and Photobiology B: Biology*, vol. 162, pp. 577-582, Sep. 2016.
- Acott et al., "Trabecular Repopulation by Anterior Trabecular Meshwork Cells After Laser Trabeculoplasty", *American Journal of Ophthalmology*, vol. 107, issue 1, pp. 1-6, Jan. 1989.
- Cao et al., "Peripheral Iridotomy," *Medscape* 25, pp. 1-12, Jun. 15, 2020.
- Husain, "Laser Peripheral Iridotomy—Practical Points", YouTube presentation, p. 1, Sep. 28, 2016, downloaded from <https://www.youtube.com/watch?v=Azxsv31yls>.
- Ivancic et al., "Early Diagnosis of Ocular Hypertension Using a Low-Intensity Laser Irradiation Test", *Photomedicine and Laser Surgery*, vol. 00, No. 00, pp. 1-5, year 2009.
- Smith et al., "Light scatter from the central human cornea", *Journal "Eye"*, issue 4, pp. 584-588, year 1990.
- Turati et al., "Patterned Laser Trabeculoplasty", *Ophthalmic Surgery, Lasers and Imaging*, vol. 41, No. 5, pp. 538-545, 2010.
- Nozaki et al., "Patterned Laser Trabeculoplasty with PASCAL streamline 577", *Investigative Ophthalmology & Visual Science*, vol. 54, p. 1867, Jun. 2013.
- Acktar, "Magic Black Coatings", Product Information, pp. 1-6, year 2017.
- Acktar, "Fractal Black Coating", Product Information, pp. 1-5, year 2017.
- Cloudy Nights LLC, "Cloudy Nights—Equipment Discussions—ATM", *Optics and DIY Forum*, pp. 1-5, Feb. 15, 2015.
- Defense Tech, "Anti-Laser Contact Lenses", Product Information, p. 1-1, Nov. 29, 2004.
- IEC standard 60825-1, "Safety of Laser Products", Edition 1.2, pp. 1-122, years 2001-2008.
- Laser Safety Industries, "Filter Specifications", pp. 1-5, year 2008.
- Thorlabs, "Laser Safety Glasses", Product Information, p. 1, Nov. 3, 2014.
- Surrey Nanosystems Ltd, "Vantablack", Data Sheet, pp. 1-4, Mar. 1, 2016.
- International Application # PCT/IB2023/061472 Search Report dated Feb. 29, 2024.
- Danielson et al., Fixed High-Energy versus Standard Titrated Energy Settings for Selective Laser Trabeculoplasty, *Journal of Glaucoma Publish Ahead of Print*, Wolters Kluwer Health, Inc., pp. 1-16, year 2023.
- Radcliffe et al., "Energy Dose-Response in Selective Laser Trabeculoplasty: A Review," *Journal of Glaucoma*, vol. 31, pp. e49-e68, year 2022.
- Gazzard, "A Brief Guide to Gonioscopy," Video Clip, *Optometry today*, pp. 1-2, May 21, 2015, as downloaded from <https://www.youtube.com/watch?v=8yTtbHWxUik>.
- Alward et al., "Principles of Gonioscopy," *Color Atlas of Gonioscopy*, American Academy of Ophthalmology, pp. 1-10, Nov. 8, 2017, as downloaded from <https://www.aao.org/education/disease-review/principles-of-gonioscopy>.

(56)

References Cited**OTHER PUBLICATIONS**

Nolan et al., "Gonioscopy skills and techniques," Community Eye Health Journal, vol. 34, No. 112, pp. 40-42, year 2021.

Breazzano et al., "Analysis of Schwalbe's Line (Limbal Smooth Zone) by Scanning Electron Microscopy and Optical Coherence Tomography in Human Eye Bank Eyes," Journal of Ophthalmic and Vision Research, vol. 8, issue 1, pp. 9-16, Jan. 2013.

Thorlabs, Inc., "CPS520—Collimated Laser Diode Module, 520 nm, 4.5 mW, Elliptical Beam, Ø11 mm," Product Details, pp. 1-1, years 1999-2023, as downloaded from <https://www.thorlabs.com/thorproduct.cfm?partnumber=CPS520>.

Prophotonix, "Green Laser Modules," Product Information, pp. 1-8, year 2024, as downloaded from <https://www.prophotonix.com/led-and-laser-products/laser-modules/laser-modules-color/green-laser-modules/>.

Idex Helath & Science Llc, "532 nm StopLine® single-notch filter," Product Details, pp. 1-2, year 2023 as downloaded from https://www.idex-hs.com/store/product-detail/nf03_532e_25/fl-009362?cat_id=products&node=individual_optical_filters.

Brackley et al., "Lecture: Using the Slit Lamp Microscope to Visualize the Ocular Structures," Video Clip, pp. 1-2, Sep. 17, 2022, as downloaded from <https://www.youtube.com/watch?v=1E-sEhy9tBo>.

Bruce et al., "Zoom in on Gonioscopy," Review of Optometry, pp. 1-8, Sep. 1, 2016, as downloaded from <https://www.reviewofoptometry.com/article/zoom-in-on-gonioscopy>.

AU Application # 2022211843 Office Action dated Jan. 8, 2024.

JP Application # 2022508451 Office Action dated Mar. 5, 2024.

AU Application # 2021369792 Office Action dated Mar. 21, 2024.

Root, "How to perform a Laser Iridotomy (Video)," pp. 1-14, year 2010, as downloaded from <https://timroot.com/how-to-perform-a-laser-iridotomy-video/>.

AU Application # 2022211843 Office Action dated Sep. 27, 2023.

AU Application # 2021311097 Office Action dated Sep. 28, 2023.

U.S. Appl. No. 17/427,926 Office Action dated Oct. 17, 2023.

JP Application # 2021536316 Office Action dated Oct. 24, 2023.

JP Application # 2020561860 Office Action dated Oct. 31, 2023.

JP Application # 2021516473 Office Action dated Nov. 7, 2023.

SG Application # 11202010437T Office Action Dec. 5, 2023.

U.S. Appl. No. 17/735,153 Office Action dated Dec. 18, 2023.

U.S. Appl. No. 17/136,052 Office Action dated Dec. 22, 2023.

U.S. Appl. No. 17/427,926 Office Action dated Dec. 22, 2023.

International Application # PCT/IB2023/060104 Search Report Dec. 26, 2023.

Sridhar, "Anatomy of Cornea and Ocular Surface," Indidan Journal of Ophthalmology, vol. 66, issue 2, pp. 190-194, year 2018.

JP Application # 2022567443 Office Action dated Dec. 17, 2024.

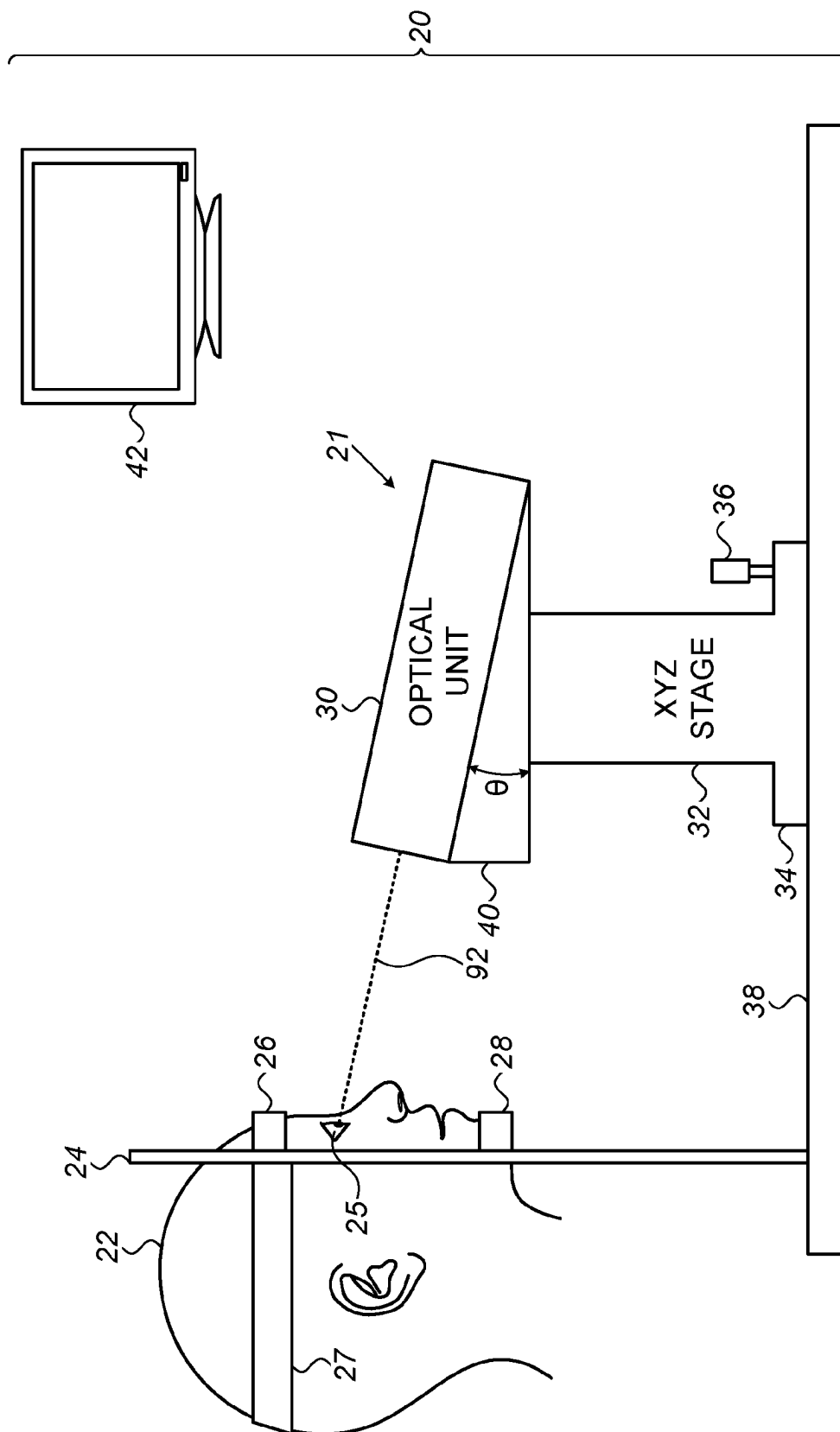
U.S. Appl. No. 17/627,701 office action dated Jul. 2, 2025.

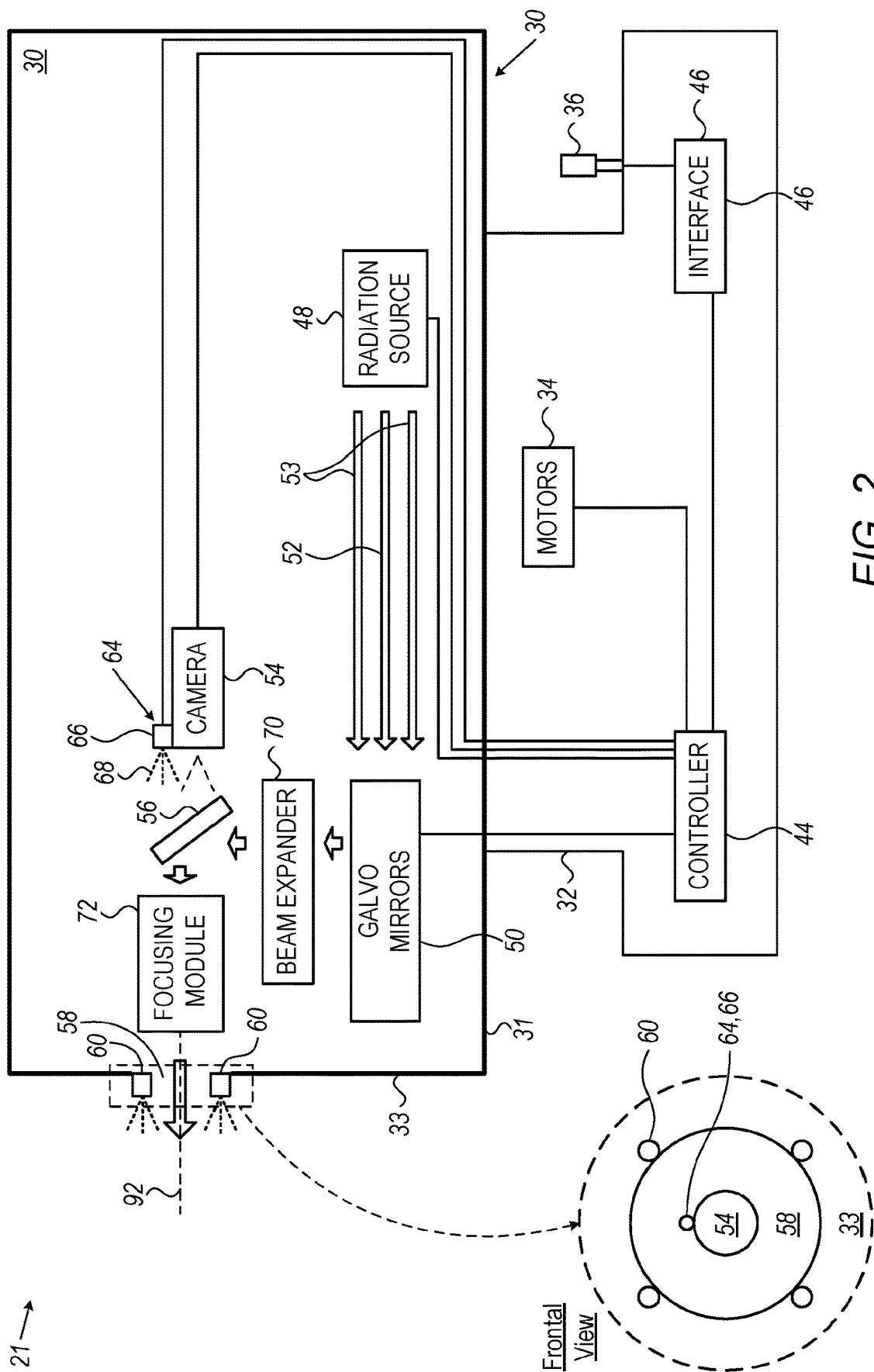
CN Application # 2021800715816 office action dated Apr. 25, 2025.

JP Application # 2022-567443 office action dated May 15, 2025.

* cited by examiner

FIG. 1





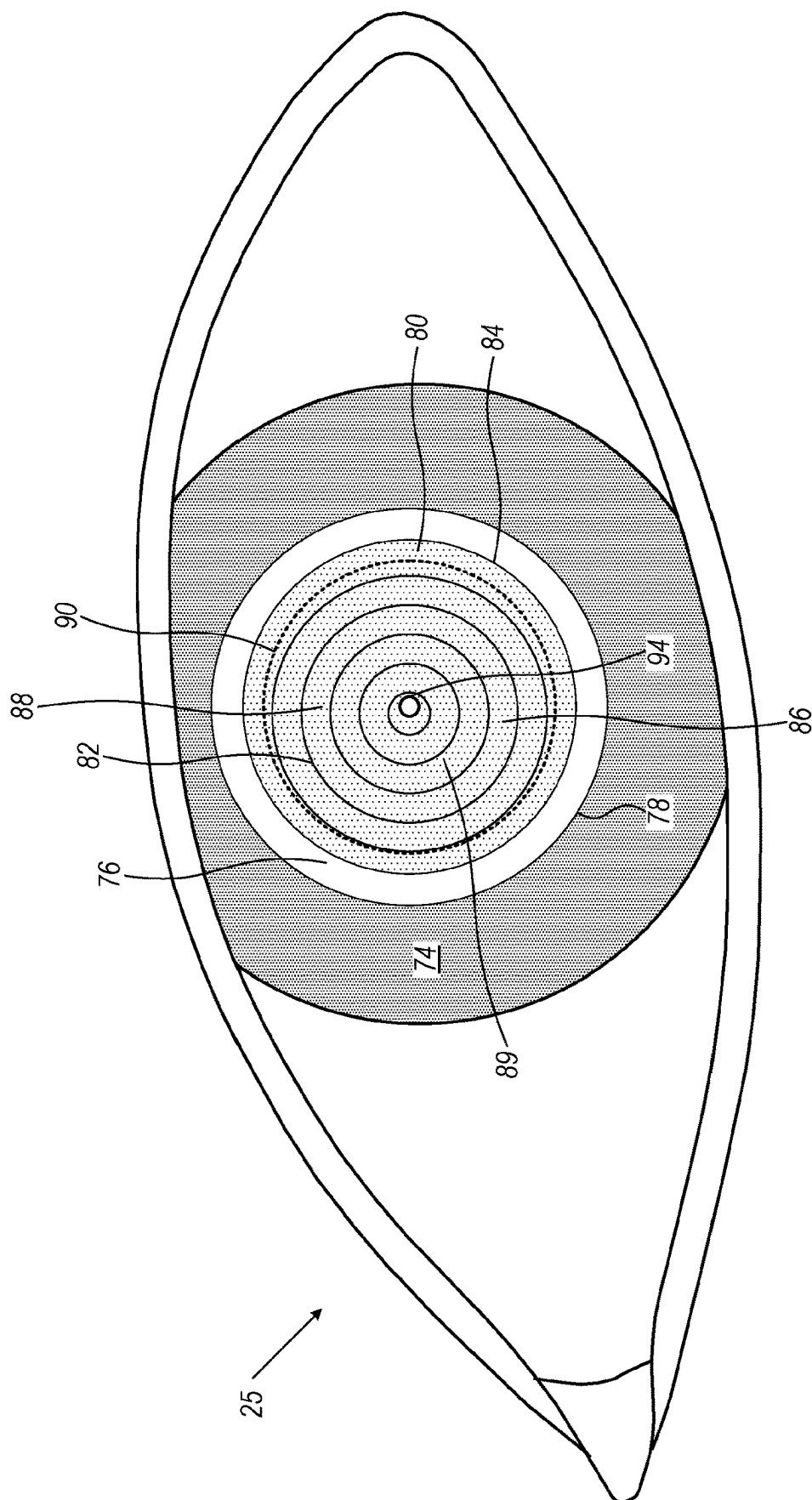


FIG. 3

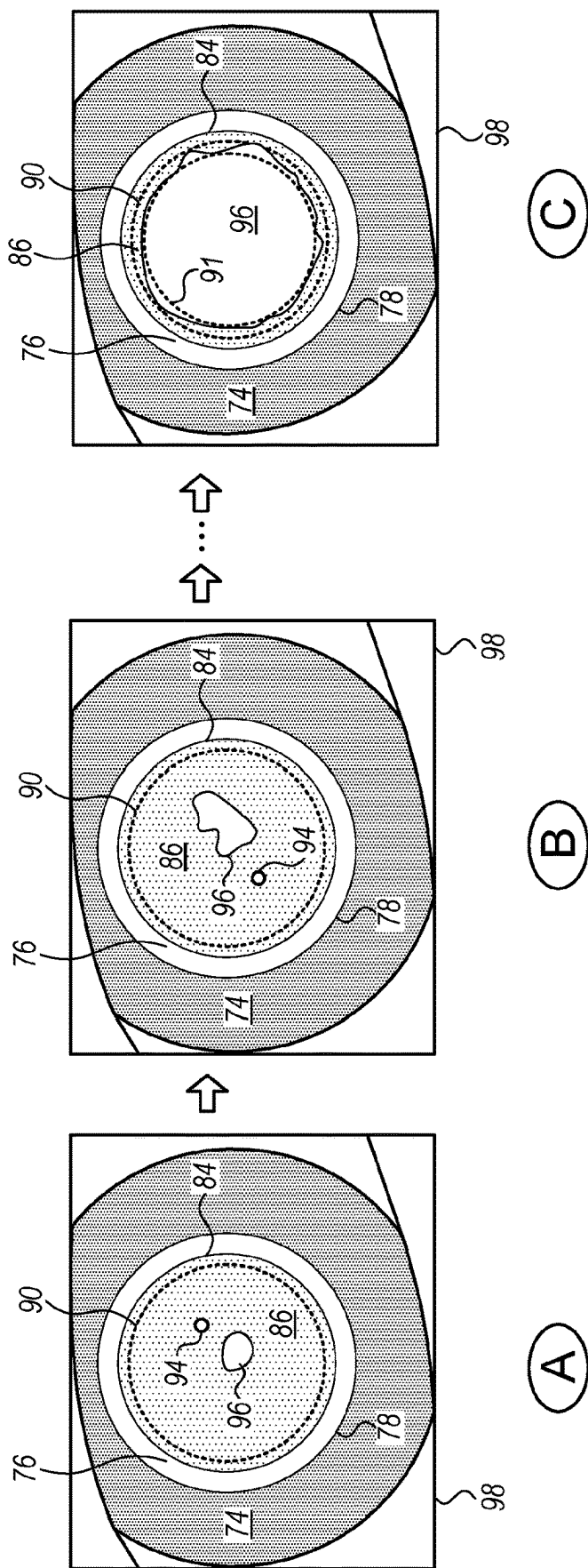


FIG. 4

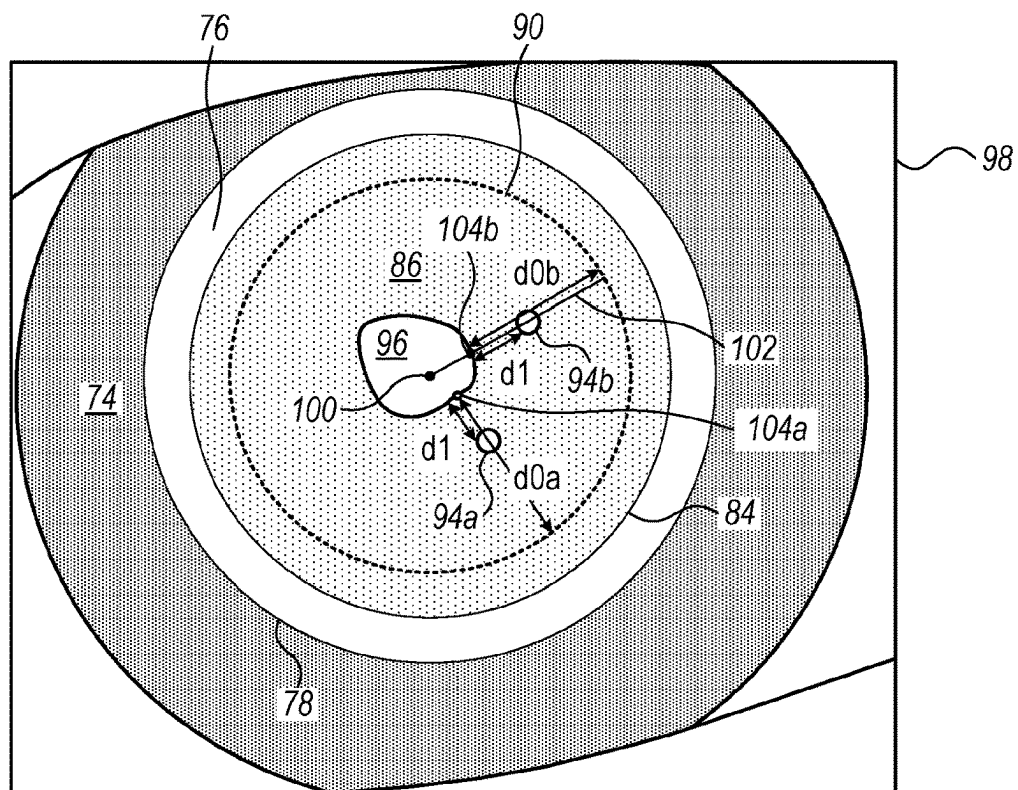


FIG. 5

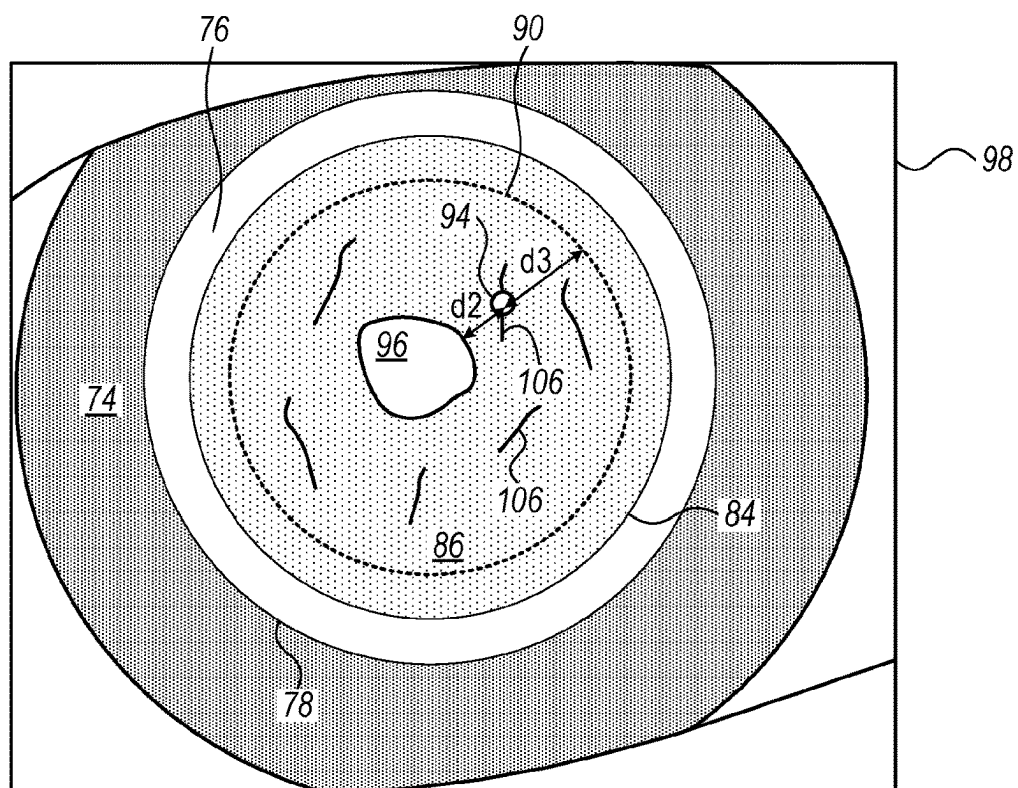


FIG. 6

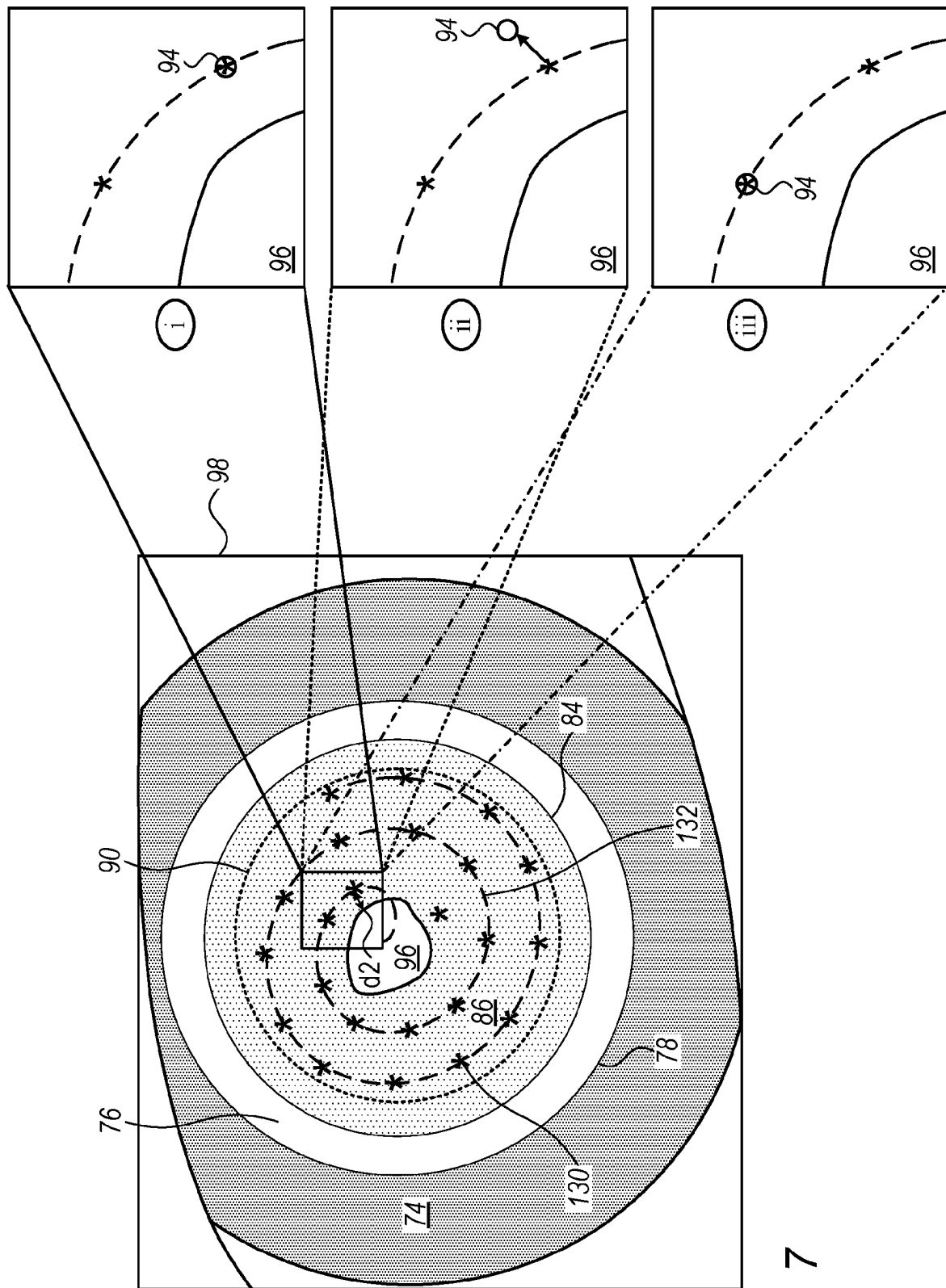


FIG. 7

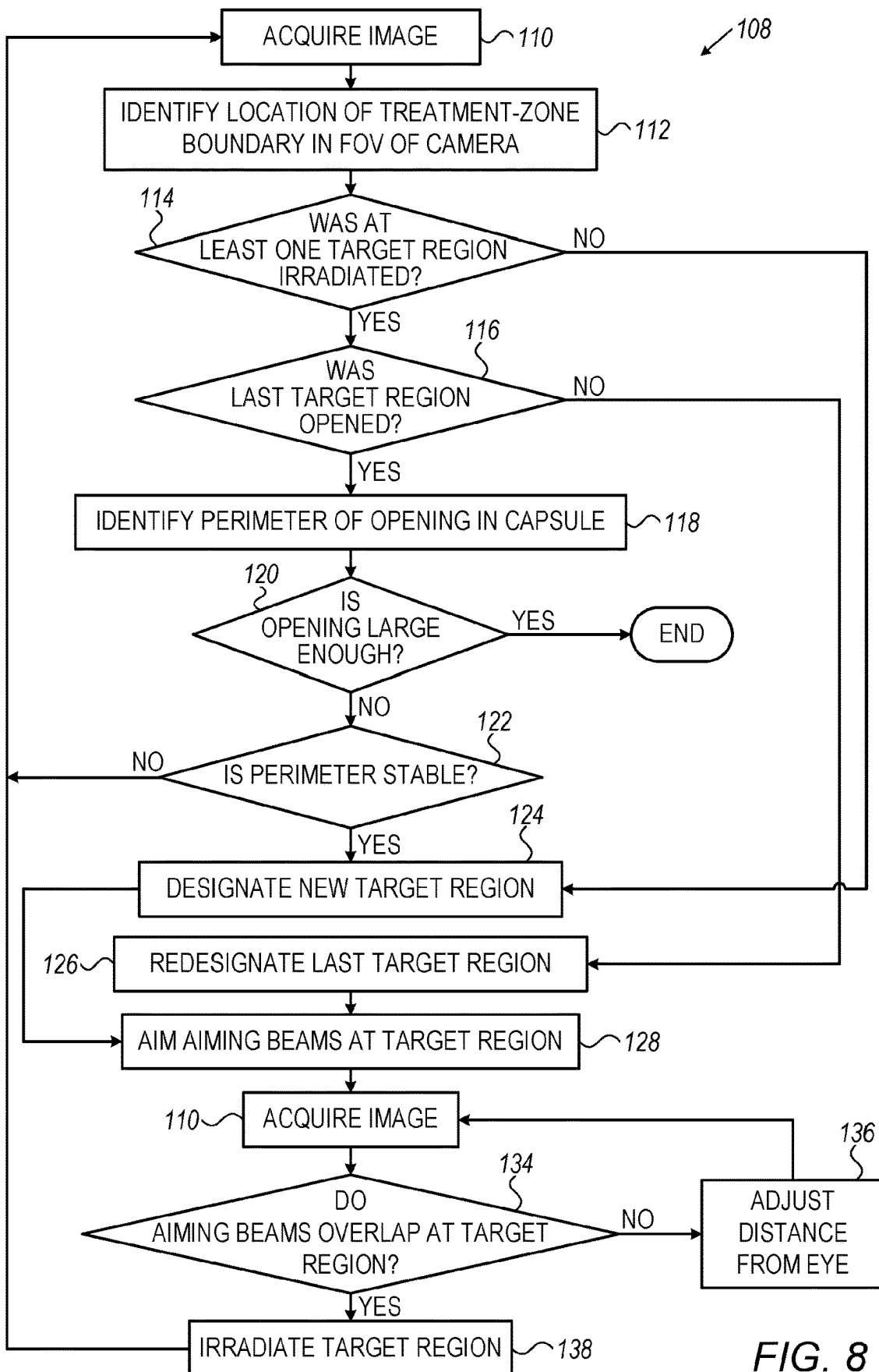


FIG. 8

1

AUTOMATED CAPSULOTOMY**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application claims the benefit of U.S. Provisional Application 63/053,650, entitled "Automatic posterior capsulotomy and combination device," filed Jul. 19, 2020, whose disclosure is incorporated herein by reference.

FIELD OF THE INVENTION

The present invention is related to ophthalmologic procedures, especially capsulotomy procedures.

BACKGROUND

In some cases, following the replacement of the natural lens of an eye, opacification of the posterior capsule of the eye occurs. In such cases, a posterior capsulotomy may be the treatment of choice.

U.S. Pat. No. 8,465,478 describes systems, apparatus, and methods for developing laser systems that can create a precise predetermined jigsaw capsulotomy. The systems, apparatus, and methods further provide laser systems that can use a single laser as a therapeutic laser and as laser radar and that reduce the patient-to-patient variability and doctor-to-doctor variability associated with hand-held apparatus for performing capsulorhexis and capsulotomies. There is further described a precise predetermined jigsaw shot pattern and shaped capsulotomy that is based at least in part on the shape of an IOL and in particular an accommodating IOL.

U.S. Pat. No. 8,845,625 describes systems and methods for cataract intervention. In one embodiment a system comprises a laser source configured to produce a treatment beam comprising a plurality of laser pulses; an integrated optical system comprising an imaging assembly operatively coupled to a treatment laser delivery assembly such that they share at least one common optical element, the integrated optical system being configured to acquire image information pertinent to one or more targeted tissue structures and direct the treatment beam in a 3-dimensional pattern to cause breakdown in at least one of the targeted tissue structures; and a controller operatively coupled to the laser source and integrated optical system, and configured to adjust the laser beam and treatment pattern based upon the image information, and distinguish two or more anatomical structures of the eye based at least in part upon a robust least squares fit analysis of the image information.

U.S. Pat. No. 10,143,590 describes a method and apparatus for performing a laser-assisted posterior capsulotomy and for performing laser eye surgery on an eye having a penetrated cornea. A method for performing a posterior capsulotomy includes injecting fluid between the lens posterior capsule and the anterior hyaloids membrane to separate the lens posterior capsule and the anterior hyaloids membrane. With the lens posterior capsule separated from the anterior hyaloids membrane, a posterior capsulotomy is performed on the lens posterior capsule by using a laser to incise the lens posterior capsule.

U.S. Pat. No. 10,849,789 describes an ophthalmic measurement and laser surgery system, including: a laser source; a corneal topography subsystem; an axis determining subsystem; a ranging subsystem comprising an Optical Coherence Tomographer (OCT); and a refractive index determining subsystem. All of the subsystems are under the operative control of a controller. The controller is configured to: operate

2

the corneal topography subsystem to obtain corneal surface information; operate the axis determining subsystem to identify one or more ophthalmic axes of the eye; operate the OCT to sequentially scan the eye in a plurality of OCT scan patterns, the plurality of scan patterns configured to determine an axial length of the eye; operate the refractive index determining subsystem so to determine an index of refraction of one or more ophthalmic tissues, wherein at least one of the corneal surface information, ophthalmic axis information, and axial length is modified based on the determined index of refraction.

SUMMARY OF THE INVENTION

There is provided, in accordance with some embodiments of the present invention, a system, including a radiation source and a controller. The controller is configured to define a treatment zone on a capsule of an eye of a subject, and to form an opening in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing the radiation source to irradiate the designated target region.

In some embodiments, the capsule is a posterior capsule.

In some embodiments, the controller is configured to define the treatment zone by:

based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye, and

defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.

In some embodiments, the controller is configured to define the treatment zone by:

based on at least one initial image of the eye, identifying an edge of an iris of the eye, and

defining the treatment zone such that the treatment zone lies entirely within the edge.

In some embodiments, the controller is configured to define the treatment zone by:

based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye,

in response to the features, calculating an estimated position of the IOL, and

defining the treatment zone in response to the estimated position.

In some embodiments, during at least one of the iterations, designating the target region includes:

based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and in response to identifying the tissue, redesignating the last-designated one of the target regions.

In some embodiments, during at least one of the iterations, designating the target region includes:

based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and

designating the target region at a predefined distance from the identified portion of the perimeter.

In some embodiments, during at least one of the iterations, designating the target region includes:

based on the image, identifying a fold within the treatment zone, and

3

in response to identifying the fold, designating the target region such that the target region overlaps the fold.

In some embodiments, during at least one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that a perimeter of the opening is stable, and

in response to the ascertaining, designating the target region.

In some embodiments,

the controller is configured to form the opening subsequently to a designation of a sequence of tentative target regions, and

during at least one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and

in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.

In some embodiments, during at least one other one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.

In some embodiments, during at least one other one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the target region by applying an offset to a location of the next one of the tentative target regions.

In some embodiments, the controller is further configured to:

define a target boundary on the capsule, and

terminate the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.

In some embodiments, the controller is configured to define the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.

There is further provided, in accordance with some embodiments of the present invention, a method, including defining a treatment zone on a capsule of an eye of a subject and, subsequently to defining the treatment zone, forming an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing a radiation source to irradiate the designated target region.

There is further provided, in accordance with some embodiments of the present invention, a computer software product including a tangible non-transitory computer-readable medium in which program instructions are stored. The instructions, when read by a controller, cause the controller to define a treatment zone on a capsule of an eye of a subject,

4

and to form an opening in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing the radiation source to irradiate the designated target region.

The present invention will be more fully understood from the following detailed description of embodiments thereof, taken together with the drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic illustration of a system for performing a capsulotomy on a capsule of an eye of a patient, in accordance with some embodiments of the present invention;

FIG. 2 is a schematic illustration of a capsulotomy device, in accordance with some embodiments of the present invention;

FIG. 3 is a schematic illustration of a method for defining a treatment zone, in accordance with some embodiments of the present invention;

FIG. 4 is a schematic illustration of the formation of an opening in a capsule, in accordance with some embodiments of the present invention;

FIGS. 5-6 are schematic illustrations of techniques for designating target regions, in accordance with some embodiments of the present invention;

FIG. 7 is a schematic illustration of a technique for designating target regions based on a predesignated sequence of tentative target regions, in accordance with some embodiments of the present invention; and

FIG. 8 is a flow diagram for an example iterative process for forming an opening in a capsule of an eye, in accordance with some embodiments of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Overview

Conventional capsulotomy procedures require a specialist physician who has undergone extensive training. Unfortunately, however, such physicians may be in short supply.

To address this challenge, embodiments of the present invention provide an automated capsulotomy system, which allows even a physician who has undergone a lesser amount of training to perform the procedure. The system comprises a radiation source, a camera, a controller, and suitable optics. Based on images acquired by the camera, the controller controls the radiation source and optics so as to form an opening in the posterior capsule by irradiating the capsule.

Typically, the controller first defines a treatment zone, which includes at least some of the opacified portion of the posterior capsule. Next, the controller designates a first target region, typically at the center of the treatment zone, and then irradiates the first target region. After confirming, from an image of the capsule, that an opening was initialized at the first target region, the controller designates a second target region at a suitable distance from the perimeter of the opening, and then irradiates the second target region. After confirming, from another image of the capsule, that the opening was expanded, the controller designates and then irradiates a third target region. In this manner, the controller uses image processing to optimize the designation of the target regions, thus facilitating expanding the opening more

efficiently, i.e., using fewer radiation beams. Upon ascertaining, from another image, that the opening has been sufficiently enlarged, the controller terminates the procedure.

Typically, the controller is configured to identify, by image processing, any folds in the posterior capsule. In response to identifying a fold, a target region overlapping the fold may be designated. Advantageously, by targeting folds in the capsule, the opening may be expanded more efficiently.

In some embodiments, a sequence of tentative target regions is designated prior to any irradiation of the capsule. Subsequently, the controller iterates through the tentative target regions. For each tentative target region, based on the proximity of the tentative target region to the perimeter of the opening, the controller decides whether to irradiate the tentative target region, skip the tentative target region, or irradiate a new target region that is offset from the tentative target region.

In addition to performing an automated capsulotomy of a posterior capsule as described herein, the controller may form slit-shaped openings in the anterior capsule of the eye. Alternatively or additionally, the controller may use any of the techniques described herein to form a larger opening in the anterior capsule.

System Description

Reference is initially made to FIG. 1, which is a schematic illustration of a system 20, comprising a capsulotomy device 21, for performing a capsulotomy on a capsule of an eye 25 of a patient 22, in accordance with some embodiments of the present invention. Reference is further made to FIG. 2, which is a schematic illustration of capsulotomy device 21, in accordance with some embodiments of the present invention.

Capsulotomy device 21 comprises an optical unit 30 and a controller 44. Optical unit 30 comprises one or more beam-directing elements, comprising, for example, one or more galvo mirrors 50, which may be referred to collectively as a “galvo scanner,” and/or a beam combiner 56. Optical unit 30 further comprises a radiation source 48, which is configured to irradiate the capsule of eye 25 with one or more treatment beams 52 by emitting the treatment beams toward the beam-directing elements such that the beams are directed by the beam-directing elements toward the capsule. Typically, radiation source 48 is further configured to irradiate the capsule with a pair of aiming beams 53, which are configured to overlap one another only when the distance of the optical unit from the capsule and the positioning of the beam-directing elements provide a desired treatment spot size on the capsule. Thus, aiming beams 53 may be used to verify the spot size of each treatment beam prior to emitting the treatment beam, as further described below with reference to FIG. 8.

In some embodiments, optical unit 30 further comprises one or more other optics. For example, the optical unit may comprise a beam expander 70, which expands and then re-collimates treatment beams 52. In such embodiments, typically, optical unit 30 further comprises a focusing module (comprising, for example, an F-theta lens or another type of lens), configured to focus the treatment beams.

More specifically, before the emission of each treatment beam 52 from radiation source 48, or while the beam is being emitted, controller 44 aims the beam-directing elements at a target region of the capsule such that the beam is directed, by the beam-directing elements, toward the target

region. For example, the beam may be deflected by galvo mirrors 50 through beam expander 70 toward beam combiner 56, and then deflected by the beam combiner through focusing module 72, such that the beam impinges on the target region with the desired spot size. (Since each treatment beam impinges on the capsule with a non-infinitesimal spot size, the present application generally describes each beam as impinging on a “region” of the capsule, rather than impinging at a “point.”) The beam thus follows a path 92, which extends from the most downstream of the beam-directing elements—such as focusing module 72—to the target region.

Typically, the radiation source comprises a laser, such as an Nd:YAG laser. (Examples of off-the-shelf products incorporating an Nd:YAG laser include Optimis II by Quantel Medical of Courmon-d’Auvergne, France, and Selecta Duet by Lumenis of Yokneam, Israel.) The laser may be modified to include an attenuator, an energy meter, and/or a mechanical shutter. Alternatively or additionally to a laser, the radiation source may comprise any other suitable emitter of radiation.

In some embodiments, the treatment beams include visible light. Alternatively or additionally, the treatment beams may include non-visible electromagnetic radiation, such as microwave radiation, infrared radiation, X-ray radiation, or gamma radiation. Typically, the wavelength of the treatment beams is between 400 and 1400 nm, e.g., 532 nm or 1064 nm.

Typically, the spatial profile of each treatment beam 52 on the capsule is approximately circular, e.g., with a spot size less than 22 μm . Alternatively, the spatial profile of each treatment beam 52 may be elliptical, square, or of any other suitable shape. In general, the size and shape of the treatment beam spots are chosen so as to deliver sufficient energy for photodisruption of the capsule.

Typically, the aiming beams include visible light, e.g., with a wavelength of between 600 and 700 nm.

Optical unit 30 further comprises a camera 54, which is used by controller 44 to acquire images of the eye. As shown in FIG. 2, camera 54 is typically aligned, at least approximately, with path 92; for example, the angle between path 92 and a hypothetical line extending from eye 25 to the camera may be less than 15 degrees. In some embodiments, the camera is positioned behind beam combiner 56, such that the camera receives light via the beam combiner. In other embodiments, the camera is offset from the beam combiner.

At the start of the procedure, camera 54 acquires an initial image of the capsule, or at least a portion thereof. Based on the image, controller 44 defines a treatment zone on the capsule, as further described below with reference to FIG. 3. Subsequently, during the procedure, camera 54 acquires multiple images of the capsule, or at least a portion thereof, at a relatively high frequency. Controller 44 processes these images and, in response thereto, designates target regions within the treatment zone for irradiation, as further described below with reference to the subsequent figures.

In general, camera 54 may comprise one or more imaging sensors of any suitable type(s), such as a charge-coupled device (CCD) sensor, a complementary metal-oxide-semiconductor (CMOS) sensor, an optical coherence tomography (OCT) sensor, and/or a hyperspectral image sensor. Using the sensors, the camera may acquire two-dimensional or three-dimensional images of any suitable type, such as monochrome images, color images (based, for example, on three color frames), multispectral images, hyperspectral

images, optical coherence tomography (OCT) images, or images produced by fusing multiple images of different respective types.

Typically, optical unit **30** further comprises a light source **66**, which is aligned, at least approximately, with path **92**. For example, the angle between path **92** and a hypothetical line extending from the end of path **92** on eye **25** to light source **66** may be less than 20 degrees, such as less than 10 degrees. Light source **66** is configured to function as a fixation target **64** by transmitting visible fixation light **68**, thus helping to stabilize the position of the eye. In particular, prior to the procedure, patient **22** is instructed to fixate eye **25** on light source **66**. Subsequently, during the procedure, by virtue of light source **66** transmitting fixation light **68**, eye **25** fixates on the light source, such that the eye's line-of-sight is approximately coincident with path **92** (due to the light source being approximately aligned with the path) and the eye is relatively stable. While the eye fixates on the light source, the radiation source irradiates the capsule of the eye with treatment beams **52**.

In some embodiments, light source **66** comprises a light emitter, such as a light emitting diode (LED). In other embodiments, the light source comprises a reflector configured to reflect light emitted from a light emitter.

Typically, the wavelength of fixation light **68**, which may be higher or lower than that of the treatment beams, is between 350 and 850 nm. For example, fixation light **68** may be orange or red, with a wavelength of 600-750 nm.

Typically, the optical unit comprises an optical bench, and at least some of the aforementioned components belonging to the optical unit, such as the radiation source, the galvo mirrors, and the beam combiner, are coupled to the optical bench. Typically, the optical unit further comprises a front face **33**, through which the treatment beams and the fixation light pass. For example, optical unit **30** may comprise an encasement **31**, which at least partially encases the optical bench and comprises front face **33**. (Encasement **31** may be made of a plastic, a metal, and/or any other suitable material.) Alternatively, front face **33** may be attached to, or may be an integral part of, the optical bench.

In some embodiments, front face **33** is shaped to define an opening **58**, through which the treatment beams and fixation light pass. In other embodiments, the front face comprises an exit window in lieu of opening **58**, such that fixation light **68** and treatment beams **52** pass through the exit window. The exit window may be made of a plastic, a glass, or any other suitable material that is generally transparent to the treatment beams, aiming beams, fixation light **68**, and any light used for imaging the eye, such as light from the illumination sources **60** described below.

Optical unit **30** is mounted onto an XYZ stage unit **32**, which is controlled by a control mechanism **36**, such as a joystick. Using control mechanism **36**, the user of system **20** may position the optical unit (e.g., by adjusting the distance of the optical unit from the eye) prior to treating the eye. In some embodiments, XYZ stage unit **32** comprises locking elements configured to inhibit motion of the stage unit following the positioning of the stage unit.

In some embodiments, XYZ stage unit **32** comprises one or more motors **34**, and control mechanism **36** is connected to interface circuitry **46**. As the user manipulates the control mechanism, interface circuitry **46** translates this activity into appropriate electronic signals, and outputs these signals to controller **44**. In response to the signals, the controller controls motors **34**.

In other embodiments, XYZ stage unit **32** is controlled manually by manipulating the control mechanism. In such embodiments, the XYZ stage unit may comprise a set of gears instead of motors **34**.

System **20** further comprises a headrest **24**, comprising a forehead rest **26** and a chinrest **28**. During the capsulotomy, patient **22** presses his forehead against forehead rest **26** while resting his chin on chinrest **28**. In some embodiments, headrest **24** further comprises an immobilization strap **27**, configured to secure the patient's head from behind and thus keep the patient's head pressed against the headrest.

In some embodiments, system **20** further comprises a contact lens, configured to focus the treatment beams and/or stabilize eye **25** while contacting the eye.

In some embodiments, as shown in FIG. 1, headrest **24** and XYZ stage unit **32** are both mounted onto a surface **38**, such as the top surface of a tray or table. In other embodiments, the XYZ stage unit is mounted onto surface **38**, and the headrest is attached to the XYZ stage unit.

System **20** further comprises a monitor **42**, configured to display the images of the eye acquired by the camera. Monitor **42** may be attached to optical unit **30** or disposed at any other suitable location, such as on surface **38** next to device **21**. In some embodiments, monitor **42** comprises a touch screen, such that the user may input commands and/or information to the system via the touch screen. Alternatively or additionally, system **20** may comprise any other suitable input devices, such as a keyboard or a mouse, which may be used by the user.

In some embodiments, monitor **42** is connected directly to controller **44** over a wired or wireless communication interface. In other embodiments, monitor **42** is connected to controller **44** via an external processor, such as a processor belonging to a standard desktop computer.

In some embodiments, as shown in FIG. 2, controller **44** is disposed within XYZ stage unit **32**. In other embodiments, controller **44** is disposed externally to the XYZ stage unit. Alternatively or additionally, the controller may cooperatively perform at least some of the functionality described herein with another, external processor.

In some embodiments, system **20** is also configured to perform trabeculoplasty procedures, e.g., as described in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference. Thus, using system **20**, a trabeculoplasty procedure may be performed on patient **22** before or after a capsulotomy procedure is performed on the patient; for example, both procedures may be performed during a single sitting of patient **22**.

In such embodiments, the treatment wavelength used for trabeculoplasty procedures (e.g., 532 nm) may be the same as that used for capsulotomy procedures. Alternatively, radiation source **48** may comprise a wavelength converter, configured to convert the wavelength of the treatment beams from a first wavelength used for capsulotomy procedures (e.g., 1064 nm) to a second wavelength used for trabeculoplasty procedures (e.g., 532 nm), or vice versa. Typically, the radiation source further comprises a switch, configured to switch the treatment beams between a first optical path, which includes the wavelength converter, and a second optical path, which does not include the wavelength converter.

Alternatively or additionally, in such embodiments, optical unit **30** may further comprise one or more illumination sources **60** comprising, for example, one or more LEDs, such as white-light or infrared LEDs. For example, the optical unit may comprise a ring of LEDs surrounding

opening **58**. In such embodiments, controller **44** may cause illumination sources **60** to intermittently flash light at the eye during a trabeculoplasty procedure, as described in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference. This flashing may facilitate the imaging performed by the camera, and, by virtue of the brightness of the flashing, may further help constrict the pupil of the eye. (For ease of illustration, the electrical connection between controller **44** and illumination sources **60** is not shown explicitly in FIG. 2.) In some embodiments, illumination sources **60** are coupled to front face **33**, as shown in FIG. 2.

As shown in FIG. 1, in some embodiments—particularly those in which a trabeculoplasty, which generally requires that the entire limbus be visible, is performed—the optical unit is directed obliquely upward toward the eye while the eye gazes obliquely downward toward the optical unit, such that path **92** is oblique. For example, the path may be oriented at an angle θ of between five and twenty degrees with respect to the horizontal. Advantageously, this orientation reduces occlusion of the patient's eye by the patient's upper eyelid and associated anatomy.

In some embodiments, as shown in FIG. 1, the oblique orientation of path **92** is achieved by virtue of the optical unit being mounted on a wedge **40**, which is mounted on the XYZ stage unit. In other words, the optical unit is mounted onto the XYZ stage unit via wedge **40**. (Wedge **40** is omitted from FIG. 2.)

In some embodiments, at least some of the functionality of controller **44**, as described herein, is implemented in hardware, e.g., using one or more fixed-function or general-purpose integrated circuits, Application-Specific Integrated Circuits (ASICs), and/or Field-Programmable Gate Arrays (FPGAs). Alternatively or additionally, controller **44** may perform at least some of the functionality described herein by executing software and/or firmware code. For example, controller **44** may be embodied as a programmed processor comprising, for example, a central processing unit (CPU) and/or a Graphics Processing Unit (GPU). Program code, including software programs, and/or data may be loaded for execution and processing by the CPU and/or GPU. The program code and/or data may be downloaded to the controller in electronic form, over a network, for example. Alternatively or additionally, the program code and/or data may be provided and/or stored on non-transitory tangible media, such as magnetic, optical, or electronic memory. Such program code and/or data, when provided to the controller, produce a machine or special-purpose computer, configured to perform the tasks described herein.

In some embodiments, the controller comprises a system on module (SOM), such as the DART-MX8M by Variscite of Lod, Israel.

Defining the Treatment Zone

Reference is now made to FIG. 3, which is a schematic illustration of a method for defining a treatment zone, in accordance with some embodiments of the present invention.

FIG. 3 shows eye **25** at the start of the capsulotomy procedure. In particular, FIG. 3 shows the anterior capsule **76** of the eye lying within the edge **78** of the iris **74** of the eye. An opening **84** has been formed in anterior capsule **76**, so as to allow the natural lens of the eye to be removed (e.g., via phacoemulsification) and a prosthetic intraocular lens (IOL) **80** to be inserted. Behind IOL **80** lies an opacified

posterior capsule **86** of the eye, on which the capsulotomy procedure is to be performed.

Typically, IOL **80** includes an elliptical optic **89** connected to curved filaments, or “haptics,” that anchor the IOL in place. For example, optic **89** may be circular, having a diameter of between 6 and 14 mm. Although IOL **80** is generally transparent, in some cases, as shown in FIG. 3, one or more features of IOL **80** (in particular, optic **89** of the IOL) may be visible behind opening **84**. For example, in some cases, optic **89** includes a Fresnel lens including multiple lens zones, and concentric circles **82**, which demarcate the lens zones, are visible.

At the start of the capsulotomy procedure, controller **44** (FIG. 2) defines a treatment zone **88**, demarcated in the present figures by a boundary **90**, on posterior capsule **86**. Typically, treatment zone **88** is elliptical, e.g., circular.

For example, based on at least one initial image of the eye acquired by camera **54** (FIG. 2), the controller may identify opening **84** in anterior capsule **76**, e.g., by applying any suitable pattern-matching and/or edge-detection techniques to the initial image. Subsequently, the controller may define the treatment zone such that the treatment zone lies entirely behind the opening. For example, the controller may offset boundary **90** a predefined distance (e.g., 0.1-1 mm) inward from the perimeter of opening **84**. As an extra safety precaution, the controller may further identify edge **78** of iris **74** based on the initial image, and verify that the treatment zone lies entirely within edge **78**.

Alternatively, based on the initial image of the eye, the controller may identify one or more features of IOL **80**, e.g., by applying any suitable pattern-matching and/or edge-detection techniques to the initial image. In response to the features, the controller may calculate an estimated position of the IOL, and then define the treatment zone in response to the estimated position. In particular, the controller may define the position and size of the treatment zone such that, even if the entire treatment zone were to be opened, the IOL would not fall through the opening. As an extra safety precaution, the controller may further verify that the treatment zone lies entirely within edge **78**, and/or verify that the treatment zone lies entirely behind opening **84**.

For example, prior to the procedure, a user may input, to the controller, relevant features of the IOL, such as the dimensions of, and/or number of lens zones belonging to, optic **89**. Subsequently, based on the input and in response to identifying one or more circles **82**, the controller may calculate the position of the center of the optic. In response thereto, the controller may center the treatment zone at the center of the optic, and/or offset boundary **90** a predefined distance (e.g., 0.5-1 mm) inward from the perimeter of the optic, e.g., per guidelines specified in the relevant medical literature.

Alternatively, the controller may define the treatment zone by offsetting boundary **90** a predefined distance (e.g., up to 1 mm) inward from edge **78**, without necessarily identifying opening **84** or any features of IOL **80**.

Typically, subsequently to defining the treatment zone, the controller overlays boundary **90**, which demarcates the treatment zone, on the initial image of the eye, and displays the image with the overlaid boundary on monitor **42** (FIG. 1). The user may then adjust the treatment zone, e.g., by using a mouse to drag one or more points on boundary **90**, prior to approving the treatment zone.

In other embodiments, the controller defines the treatment zone in response to input from the user, without necessarily performing any of the image processing described above.

11

For example, the user may input the desired center and radius of the treatment zone, and/or drag points on boundary 90 as described above.

Forming the Opening

Subsequently to defining treatment zone 88, the controller forms an opening in capsule 86 by irradiating multiple target regions 94 within the treatment zone in an iterative process. During each iteration of the process, the controller acquires an image of at least part of the capsule, designates one of the target regions 94 based on the acquired image, and causes radiation source 48 (FIG. 2) to irradiate the designated target region. For example, FIG. 3 shows an initial target region at the center of treatment zone 88, which the controller may designate and then irradiate during the first iteration of the process.

For further details, reference is now made to FIG. 4, which is a schematic illustration of the formation of an opening 96 in capsule 86, in accordance with some embodiments of the present invention.

Section A of FIG. 4 shows an image 98 of the eye following the initialization of opening 96, which is performed by irradiating the first designated target region 94 shown in FIG. 3. Based on the image, the controller designates a second target region 94, e.g., as further described below with reference to FIG. 5.

Section B of FIG. 4 shows another image 98 following the irradiation of the second target region, which expands opening 96. Based on the image, the controller designates a third target region 94.

Section C of FIG. 4 shows another image 98 following a number of further iterations in which target regions 94 are irradiated. Based on such an image, the controller may decide to refrain from designating any further target regions, thus terminating the iterative process.

For example, the controller may define a target boundary 91 for the opening, e.g., by placing target boundary 91 at a predefined offset (e.g., 0.01-1 mm) inward from boundary 90 of the treatment zone. Subsequently, the controller may terminate the process in response to ascertaining that the opening reaches a predefined threshold percentage (e.g., 90%-100%) of target boundary 91. (In such embodiments, prior to the first treatment beam, target boundary 91 may be overlaid on the initial image of the eye and the user may then adjust the target boundary, as described above for boundary 90 with reference to FIG. 3.) Alternatively (e.g., if the pupil of the eye is relatively small), the controller may terminate the process in response to ascertaining that the opening reaches a predefined threshold percentage (e.g., 90%-100%) of boundary 90 of the treatment zone.

Reference is now made to FIG. 5, which is a schematic illustration of a technique for designating target regions 94, in accordance with some embodiments of the present invention.

FIG. 5 shows the scenario generally shown in Sections A-B of FIG. 4, in which opening 96 is being formed. In this scenario, in some cases, the controller designates the next target region by performing the following technique.

First, based on image 98, the controller identifies a portion of the perimeter of the opening that is farther from boundary 90 than are other portions of the perimeter.

For example, for multiple points 104a along the perimeter, the controller may calculate the shortest distance d0a between point 104a and boundary 90, i.e., the length of the

12

shortest line between point 104a and boundary 90. Subsequently, the controller may identify the point 104a for which d0a is a maximum.

Alternatively, for multiple angles, the controller may calculate the distance d0b between (i) the point 104b at which a line 102 radiating from the center 100 of the treatment zone at the angle intersects the perimeter of opening 96, and (ii) the point at which line 102 intersects boundary 90. Subsequently, the controller may identify the point 104b for which d0b is a maximum.

Next, the controller designates target region 94 at a predefined distance d1 from the identified portion of the perimeter. For example, subsequently to identifying a point 104a as described above, the controller may designate a target region 94a lying on the shortest line between the identified point and boundary 90 at a distance of d1 from the identified point 104a. Similarly, subsequently to identifying a point 104b as described above, the controller may designate a target region 94b lying on the line 102 joining the identified point to boundary 90, at a distance d1 from the identified point 104b.

Reference is now made to FIG. 6, which is a schematic illustration of another technique for designating target region 94, in accordance with some embodiments of the present invention.

In some cases, at least one fold 106 in the tissue of the capsule lies within the treatment zone. In such cases, the controller may identify fold 106, e.g., by applying an edge-detection algorithm, a line-detection algorithm, or any other suitable image-processing algorithm to image 98. Subsequently, in response to identifying the fold, the controller may designate target region 94 such that the target region overlaps the fold.

For example, for each of the identified folds, the controller may calculate two distances: (i) the distance d2 between the center of the fold and the nearest point on the perimeter of the opening, and (ii) the distance d3 between the center of the fold and the nearest point on boundary 90. The controller may then identify a set of "targetable" folds for which d2 is within a predefined range of distances (e.g., 0.2-2 mm, such as 0.5-1 mm). Subsequently, the controller may select the targetable fold for which d3 is a maximum, and then place target region 94 at the center of the selected fold. Alternatively, if there are no targetable folds, the controller may instead use the technique of FIG. 5 to designate the target region.

Reference is now made to FIG. 7, which is a schematic illustration of a technique for designating target regions based on a predesignated sequence of tentative target regions, in accordance with some embodiments of the present invention.

In some embodiments, prior to irradiating the capsule (and subsequently to the definition of the treatment zone), a sequence of tentative target regions 130 is designated. For example, the controller may display an image of the capsule to the user, and the user may then manually designate the sequence by clicking a mouse over each portion of the image at which a tentative target region 130 is desired. Alternatively, the controller may designate the sequence and display the sequence to the user, and the user may then approve the sequence after shifting, adding, or deleting any number of tentative target regions 130.

Typically, the sequence generally follows a geometric pattern. For example, the tentative target regions may be arranged in a spiral 132, which spirals outward from the center of the treatment zone, or in multiple crossed lines that define an asterisk. Parts of the sequence may deviate from

the pattern slightly, e.g., so as to allow one or more tentative target regions **130** to overlap folds **106** (FIG. 6).

In such embodiments, when forming opening **96**, the controller iterates through the sequence of tentative target regions **130**. For each of the tentative target regions, the controller either (i) designates the tentative target region as the next target region **94**, i.e., approves the tentative target region for irradiation, as illustrated in the top-right inset portion of FIG. 7, (ii) designates the next target region by applying an offset (typically away from the perimeter of opening **96**) to the location of the tentative target region, as illustrated in the middle-right inset portion of FIG. 7, or (iii) designates the following tentative target region as the next target region, as illustrated in the bottom-right inset portion of FIG. 7.

In particular, for each tentative target region, the controller ascertains whether the distance **d2** between the tentative target region and the perimeter of opening **96** is greater than a predefined threshold distance (e.g., 0.2-0.9 mm). If **d2** is greater than the predefined threshold distance, the tentative target region is designated as the target region **94**. Otherwise, the controller either applies the offset or designates the following tentative target region.

Notwithstanding the examples shown in FIGS. 4-7, it is noted that opening **96** need not necessarily be initialized at the center of the treatment zone. For example, when sweeping across the capsule in a cross or asterisk pattern, the first target region in each line of the cross or asterisk may be designated near the edge of the treatment zone.

Example Iterative Process

Reference is now made to FIG. 8, which is a flow diagram for an example iterative process **108** for forming an opening in a capsule of an eye, in accordance with some embodiments of the present invention. Iterative process **108** may be performed by the controller subsequently to defining the treatment zone, e.g., as described above with reference to FIG. 3.

Each iteration of process **108** begins with an imaging step **110**, at which the controller, using camera **54** (FIG. 2), acquires an image of at least part of the capsule. Subsequently to imaging step **110**, the controller, at a location-identifying step **112**, identifies the location of boundary **90** (FIG. 3) of the treatment zone in the field of view (FOV) of the camera, by repeating whichever image processing was used previously to define the treatment zone. (The location of the treatment-zone boundary in the FOV may change throughout the procedure, due to movement of the eye.)

For example, if the treatment zone was defined by placing boundary **90** at a predefined distance from the perimeter of opening **84** (FIG. 3), the controller may identify the perimeter of opening **84** in the latest acquired image, and then identify the location of the treatment-zone boundary at the predefined distance therefrom. For embodiments in which target boundary **91** (FIG. 4) is defined, the controller may also identify the location of boundary **91** in the FOV of the camera.

Subsequently, at a checking step **114**, the controller checks whether at least one target region was already irradiated (i.e., whether at least one iteration was already performed). If not, the controller performs a target-designating step **124**, described below. Otherwise, the controller checks, at another checking step **116**, whether the last-designated (and irradiated) target region was opened. In

other words, based on the acquired image, the controller checks whether tissue of the capsule is present at the last-designated target region.

Typically, checking step **116** comprises two sub-steps. In the first sub-step, the controller identifies the location of the last target region based on the location of boundary **90**. For example, based on the location of boundary **90**, the controller may calculate any shift in the location of the center of the treatment zone, and then apply the same shift to the location of the last target region. Thus, for example, assuming that (i) the last target region was centered at (x0,y0) with the center of the treatment zone being at (0,0), and (ii) the center of the treatment zone is now at (dx,dy), the controller may identify (x0+dx,y0+dy) as the new location of the last target region. Subsequently, in the second sub-step, the controller computes the change in pixel value between the last target region in the current image and the last target region in the last image, and compares this change to a predefined threshold.

If the last target region was not opened (i.e., if the controller identifies capsular tissue at the last target region), the controller, at a target-redesignating step **126**, redesignates the last target region. In some embodiments, the controller may also increase, by a predefined increment, the energy with which the target region is subsequently irradiated. (In such embodiments, typically, the energy may be increased multiple times by the predefined increment up to a predefined maximum energy, after which, if the target region is still not opened, process **108** is aborted.)

Alternatively, if the last target region was opened, the controller, at a perimeter-identifying step **118**, identifies the perimeter of opening **96** (FIG. 4), e.g., using any suitable edge-detection algorithm. Subsequently, at a size-checking step **120**, the controller checks whether the opening is large enough, e.g., by checking whether the opening reaches target boundary **91** as described above with reference to FIG. 4. If yes, process **108** ends. Otherwise, the controller proceeds to a stability-checking step **122**.

At stability-checking step **122**, the controller checks the stability of the perimeter of the opening by comparing the current location of the perimeter relative to the treatment-zone boundary, and shape of the perimeter, to the last location and shape. In response to ascertaining that the perimeter is stable, the controller proceeds to target-designating step **124**. Alternatively, in response to ascertaining that the perimeter is not stable—i.e., that the opening is still expanding as a result of previous irradiation—the controller refrains from designating a new target region, and returns to imaging step **110**.

At target-designating step **124**, the controller designates a new target region. For example, the controller may first check for any targetable folds, as described above with reference to FIG. 6. If a targetable fold is found, the controller may target the fold, as further described above with reference to FIG. 6. Otherwise, the controller may designate a target region as described above with reference to FIG. 5 or FIG. 7.

Subsequently to performing target-designating step **124** or target-redesignating step **126**, the controller, at an aiming step **128**, aims aiming beams **53** (FIG. 2) at the designated target region. Subsequently, the controller acquires another image of the capsule at imaging step **110**. The controller then checks, at a checking step **134**, whether the aiming beams overlap one another at the designated target region. If not, the controller, at a distance-adjusting step **136**, adjusts the distance of optical unit **30** from the eye, e.g., by communicating appropriate control signals to motors **34** (FIG. 2). Alternatively or additionally, the controller may adjust the

15

position of at least one optic in the optical unit, such as galvo mirrors **50** or focusing module **72** (FIG. **2**). The controller then acquires another image of the capsule, and repeats checking step **134**. In this manner, the distance of the optical unit from the eye, and/or the position of the optic, may be iteratively adjusted by the controller. Alternatively, the distance of the optical unit from the eye may be manually adjusted by the user.

Upon the controller (or the user) ascertaining that the aiming beams overlap one another, the controller causes radiation source **48** (FIG. **2**) to irradiate the designated target region, at an irradiating step **138**. The controller then begins another iteration of process **108**.

In some embodiments, the aiming beams are shaped to define different respective portions of a predefined composite pattern, such that the predefined composite pattern is formed on the capsule only when the aiming beams overlap one another. Examples of such patterns are described, for example, in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference.

It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

The invention claimed is:

1. A system, comprising:
 - a radiation source; and
 - a controller, configured to:
 - define a treatment zone on a capsule of an eye of a subject, and
 - subsequently to defining the treatment zone, form an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process:
 - acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions on the at least part of the capsule, and
 - causing the radiation source to irradiate the designated target region.
2. The system according to claim 1, wherein the capsule is a posterior capsule.
3. The system according to claim 2, wherein the controller is configured to define the treatment zone by:
 - based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye, and
 - defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.
4. The system according to claim 1, wherein the controller is configured to define the treatment zone by:
 - based on at least one initial image of the eye, identifying an edge of an iris of the eye, and
 - defining the treatment zone such that the treatment zone lies entirely within the edge.
5. The system according to claim 1, wherein the controller is configured to define the treatment zone by:
 - based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye,

16

in response to the features, calculating an estimated position of the IOL, and
defining the treatment zone in response to the estimated position.

6. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes:

- based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and

- in response to identifying the tissue, redesignating the last-designated one of the target regions.

7. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes:

- based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and

- designating the target region at a predefined distance from the identified portion of the perimeter.

8. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes:

- based on the image, identifying a fold within the treatment zone, and

- in response to identifying the fold, designating the target region such that the target region overlaps the fold.

9. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes:

- based on the acquired image, ascertaining that a perimeter of the opening is stable, and

- in response to the ascertaining, designating the target region.

10. The system according to claim 1,

- wherein the controller is configured to form the opening subsequently to a designation of a sequence of tentative target regions, and

- wherein, during at least one of the iterations, designating the target region includes:

- based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and

- in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.

11. The system according to claim 10, wherein, during at least one other one of the iterations, designating the target region includes:

- based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

- in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.

12. The system according to claim 10, wherein, during at least one other one of the iterations, designating the target region includes:

- based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

- in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the

17

target region by applying an offset to a location of the next one of the tentative target regions.

13. The system according to claim 1, wherein the controller is further configured to:

define a target boundary on the capsule, and
 terminate the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.

14. The system according to claim 13, wherein the controller is configured to define the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.

15. A method, comprising:

defining a treatment zone on a capsule of an eye of a subject; and

subsequently to defining the treatment zone, forming an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process:

acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions on the at least part of the capsule, and causing a radiation source to irradiate the designated target region.

16. The method according to claim 15, wherein the capsule is a posterior capsule.

17. The method according to claim 16, wherein defining the treatment zone comprises:

based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye; and

defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.

18. The method according to claim 15, wherein defining the treatment zone comprises:

based on at least one initial image of the eye, identifying an edge of an iris of the eye; and

defining the treatment zone such that the treatment zone lies entirely within the edge.

19. The method according to claim 15, wherein defining the treatment zone comprises:

based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye;

in response to the features, calculating an estimated position of the IOL; and

defining the treatment zone in response to the estimated position.

20. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes:

based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and

in response to identifying the tissue, redesignating the last-designated one of the target regions.

21. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes:

based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and

designating the target region at a predefined distance from the identified portion of the perimeter.

18

22. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes:

based on the image, identifying a fold within the treatment zone, and

in response to identifying the fold, designating the target region such that the target region overlaps the fold.

23. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that a perimeter of the opening is stable, and

in response to the ascertaining, designating the target region.

24. The method according to claim 15,

wherein forming the opening comprises forming the opening subsequently to a designation of a sequence of tentative target regions, and

wherein, during at least one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and

in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.

25. The method according to claim 24, wherein, during at least one other one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.

26. The method according to claim 24, wherein, during at least one other one of the iterations, designating the target region includes:

based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and

in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the target region by applying an offset to a location of the next one of the tentative target regions.

27. The method according to claim 15, further comprising:

defining a target boundary on the capsule; and terminating the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.

28. The method according to claim 27, wherein defining the target boundary comprises defining the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.

29. A computer software product comprising a tangible non-transitory computer-readable medium in which program instructions are stored, which instructions, when read by a controller, cause the controller to:

define a treatment zone on a capsule of an eye of a subject, and

subsequently to defining the treatment zone, form an opening in the capsule by irradiating multiple target

regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process:

acquiring an image of at least part of the capsule,
based on the acquired image, designating one of the 5
target regions on the at least part of the capsule, and
causing a radiation source to irradiate the designated
target region.

* * * * *